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Effects of information search tasks on the comprehension of instructional text

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Effects of information search tasks on the comprehension of instructional textAbstract

Two experiments investigated the effects of information search tasks on University students' comprehension of a scientific text. In both experiments, undergraduate students read a 2000-word introductory text on atomic models; then they searched through the text in order to answer a series of high level or low level questions, and finally they wrote a one-page summary of the text. In experiment 1, high level questions promoted a review-and-integrate search pattern, while low level questions triggered a locate-and-memorize pattern. Moreover the students tended to include information relevant to the search questions in their summary. In experiment 2 a structured overview (table of contents) increased the initial (pre-search) study time, facilitated information localization but did not influence the summary. Again, students included question-relevant materials in their summaries. It is concluded that text processing can be successfully guided toward deeper levels of comprehension, by means of appropriate task settings and presentation formats.

Effects of information search tasks on the comprehension of instructional text

The comprehension of complex instructional texts requires elaborate study strategies. According to van Dijk & Kintsch (1983), comprehension is achieved when the reader builds a mental representation of the situation described in the text, or situation model. The construction of a situation model involves several sub-processes, such as parsing the literal linguistic content, constructing micro- and macro- semantic propositions, generating various kinds of inferences and integrating text information with one's previous knowledge (Kintsch, 1998; Kintsch & van Dijk, 1978). The precise nature of comprehension processes (e.g., the extent to which readers generate specific types of inferences) depends on the reader's comprehension strategy (van Dijk & Kintsch, 1983). Most common reading situations involve "basic" strategies based on referential and causal coherence (Kintsch & van Dijk, 1978; Trabasso & van den Broek, 1985; Van den Broek, Young, Tzeng & Linderholm, 1999).

When reading lengthy texts for specific study purposes, however, more elaborate strategies may be warranted. Students must build up explicit study goals and fulfill them using relevant text information. For instance, when studying a physics text describing the evolution of atomic models, the reader needs to construct a representation of each model, then to understand the problems of the model and the features of a better model that replaced it. Even though referential and causal coherence strategies may still apply (e.g., to comprehend the relationships between empirical evidence and a particular atomic model), different strategies are needed in order to capture the global meaning of the text (e.g., comparing features across models). Other study tasks that require elaborate strategies include learning from multiple historical documents (Rouet, Britt, Mason & Perfetti, 1996; Wiley & Voss, 1999), or searching for specific information in textbook chapters (Dreher & Guthrie, 1990). There is evidence that even mature readers

experience some difficulty in building and applying elaborate comprehension strategies (Lonka, Lindblom-Ylänne & Maury, 1994; Rouet, Favart, Britt & Perfetti, 1997). It is an issue of both theoretical and practical interest to find out if students' comprehension strategies can be improved through appropriate task settings.

The purpose of this study is to examine the effect of different types of search tasks on undergraduate students' information-seeking strategies in a scientific text and on the organization of their mental representation of the text, as assessed in a summary task. We start with a brief review of the role of adjunct questions in learning from text. Then we examine in greater detail the cognitive processes involved in answering questions from text, as well as the role of individual and text presentation variables on the effectiveness of search strategies.

The role of adjunct questions in text comprehension: Managing the reader's purposes.

There is ample evidence that study directions, adjunct questions or instructional objectives deeply influence the processing of instructional materials (see Andre, 1979; Hamilton, 1985; Hartley & Davies, 1976; Rickards, 1979 for reviews). Rickards elicited two main dimensions of inserted questions: The direction of their effect (forward or backward) and the kind of processing they induced (general or specific). Backward questions encourage the mental review of the previously read text, while forward questions focus students' attention on specific information in the upcoming portion of the text (Rothkopf & Bisbicos, 1967). General questions encourage the mental review of several pieces of information while specific questions increase memory for a unique piece of information.

Backward questions may be used as a guide for students to review important information. Andre and Thieman (1988) found that students who reviewed the text in order to check the correctness of their answers obtained better scores on the same questions repeated later, compared

to students who received external feedback or no feedback on their answers. There is also some evidence that simple, well-structured search tasks may be of some help to less efficient learners in the elementary grades. Cataldo and Cornoldi (1998) found that when 6th and 7th grade poor comprehenders were asked to locate and underline relevant information in the text before actually answering a set of comprehension questions, they later obtained better scores on a comprehension post-test.

General, or "higher-level" questions facilitate deep comprehension, i.e., the integration of text elements and the production of inferences. Andre (1979) argued that higher-level questions encourage readers to pay attention to broader portions of the text and thus help them structure the mental representation of the text. Wixson (1983) found that textually explicit, textually-implicit or schema-based questions each promoted the corresponding type of inferences in 10th graders, as evidenced in a delayed recall task. Vidal-Abarca, Mengual, Sanjose and Rouet (1996) asked high-school students to study a physics text, then to review it in order to solve either text-explicit or inference questions. The students who received inference questions scored better on recall and inferential comprehension tests. However, these results were only partially replicated in another study (Vidal-Abarca, Gilabert & Rouet, 1998), possibly because the latter study involved students with much more prior knowledge of the topic.

Thus, there is ample evidence that inserted questions facilitate comprehension, especially if they promote the production of inferences and/or the integration of text elements. However, there is currently no comprehensive processing framework to account for those effects.

The cognitive processes of text-based question answering

Answering questions based on text information requires two types of processes: Memory search processes and text search processes. Graesser and Franklin (1990) proposed a cognitive

model of question answering from memory. According to the QUEST model, question answering involves a categorization mechanism, that identifies the type of question (e.g., why-, how- type of question), its focus (e.g., "water is heated" in "How is water heated?"), and the relevant information sources, that is, the episodic or general knowledge structures which may be tapped for answers to the question. Simple questions focus on a single concept or a semantic proposition. However, more complex questions focus on broader conceptual structures. The actual process of answering a question starts with the activation of a knowledge node, either directly if the node matches a term in the question, or indirectly through the contextual activation of relevant knowledge structures (see also Whilite, 1985). Search continues through a radiation mechanism called "arc-search procedure". Arc-search allows the propagation of activation through a knowledge network, based on constraints specific to each type of question. For instance, how-questions call for the search of causal antecedents or subordinate goals. Arc-search reduces the search space by identifying those knowledge nodes that may be chosen for an answer. The search space is further reduced by constraints propagation and pragmatic rules.

When answering questions from text, the processes of categorization and focusing also apply, as well as the initial activation of knowledge nodes. However the actual search process includes text inspection in addition to memory arc-search. Mosenthal (1996) identified four variables that influence text inspection: (a) document complexity is a function of the structural organization of the document (e.g., simple list vs. nested list) and the amount of raw information included in the document; (b) the type of information requested varies from the most concrete (e.g., identifying a person or a place) to the most abstract (e.g., identifying equivalence or difference between trends); (c) the type of match, a complex aggregated variable that encompasses both the type of search strategy (e.g., locate a single piece vs. integrate several

pieces of information) and the amount and type of inferencing required to match document information with requested information (no inference, low-text inference, high-text inference); and (d) plausibility of distractors, which depends on the presence and location of information that shares some features with the requested information. Based on the analysis of 217 document search tasks taken from 5 national surveys on adult document literacy, Mosenthal showed that the four variables accounted for 80% of the variance in task difficulty. Thus, when searching an external source, the subject's strategy is strongly influenced by the amount of information to be searched, its internal organization, and the relationship between the search question and the nature of the information found in the external source.

Differential effects of high versus low level questions

The processes involved in answering high- and low level questions from text differ in several respects. High level questions are usually defined as questions that focus on a broader set of concepts, or on concepts higher in a hierarchy (Andre, 1979; Whilite, 1985; Halpain, Glover & Harvey, 1985). Therefore, answering such questions requires the activation of a larger number of knowledge nodes, as well as the (re) processing of more text units. In addition, high level questions require the students to build up more connections between knowledge and text information. Such connections are established through inferencing and the restructuring of previous knowledge. Thus, the role of high level questions is similar to that of self-explanation (Chi, de Leeuw, Chiu & LaVander, 1994). In contrast, low level questions focus on a single concept and require only superficial processing of the materials. They do not provide the reader with an opportunity to generate inferences or to reorganize their previous knowledge.

The processing differences between high and low level questions are further assessed by measures of their respective attentional demands. Halpain, Glover and Harvey (1985) conducted

five experiments in order to study the influence of question type on the allocation of cognitive resources during reading. They presented University students a 1200-word expository text, in which high- and low level questions dealing with the contents of each paragraph were inserted. Using a secondary task method, Halpain et al. found that reading paragraphs preceded by high level questions required more cognitive resources than reading paragraphs preceded by low level questions. The authors also found that when questions were presented after the corresponding paragraphs, reading high level questions took more resources than reading low level questions. The difference disappeared when questions were presented before the relevant paragraphs. Thus, when the students did not have the opportunity to read the text prior to the questions, the amount of resources devoted to high and low questions did not differ. These findings confirm the view that high level postquestions require the reader to review and integrate several text elements, creating more connections in long-term memory. According to this view, the process of answering text-based high level questions is more resource consuming, but promotes deep comprehension.

Difference in answering high vs. low level questions is apparent in the pattern of search of information to answer either one or the other type of questions. Vidal-Abarca et al. (1998) conducted two experiments in which high-school (experiment 1) and University students (experiment 2) read a 2500-word physics text presented paragraph-by-paragraph on a computer screen. Then, half of the students in each experiment answered text-based explicit questions whereas the other half answered global and inference questions. In both conditions, the students were allowed to search information in the text to answer the questions. Answering both types of questions required that students focus on the same textual information. Students who answered explicit questions searched fewer number of paragraphs per question than those answering global

and inference questions. Thus, answering explicit question made students reread isolated pieces of information, whereas answering global and inference questions made them review and integrate information in different paragraphs. Another interesting result found by Vidal-Abarca et al. was that students who answered global and inference questions significantly spent less time searching information than students who answered text-based explicit questions.

The main purpose of our studies was to provide direct evidence of the differential effects of high vs. low level questions on students' text reviewing patterns. We hypothesized that searching the text with the purpose of answering higher level questions would require students to locate, review and integrate several sections of the passage, whereas low level questions would only require students to locate specific information corresponding to a narrow focus. Consequently, high level questions would improve students' memory for the text's main point. This hypothesis was examined in experiment 1. In experiment 2, we examined the influence of a structured overview on students' search strategy and comprehension. The rationale was that the overview could help students locate information for both types of search tasks. The facilitation would result in a better structured representation for students searching higher-level information. The two experiments used an on-line text presentation software which recorded subjects' reading and search behavior. The main research questions were the following: Do college-level students adjust their search patterns as a function of question type? Does the type of search questions influence what students consider "main points" in a text? And does the effect of search tasks depend on the presence of a structured overview?

Experiment 1

Rationale

Experiment 1 examined the effects of different sets of questions on students' search behavior and on their understanding of an introductory text on atomic models. Based on previous work by Vidal-Abarca et al. (1996, 1998), we expected that high level questions would lead students to review larger portions of the text, and to generate more knowledge-based inferences. The extra processing effort would result in improved memory for gist information. In contrast, low level questions would result in a locate-and-memorize type of strategy, with little effect on students' main idea comprehension.

Method

Participants: Ten University of Poitiers students (4 females and 6 males) between 18 and 25 years of age participated individually as paid volunteers. All the participants were native French speakers. Three participants (2 females, 1 male) were enrolled in non scientific undergraduate programs, the seven others were enrolled in science programs. None of the students was taking any course on atomic models at the time of the experiment.

Materials: We adapted a 2000 word-long scientific text used in previous studies (Vidal-Abarca et al., 1996)¹. The text presented three different atomic models (Dalton's, Thomson's, and Rutherford's models, respectively), as well as the theoretical and empirical reasons for replacing older models with new ones. The text was presented on a computer screen using the Select-the-Text software developed at the Learning Technology Center at Vanderbilt University (Goldman & Saul, 1990). Select-the-Text allows the computerized presentation of text in a fashion similar to that of a word processor. The text is presented one page at a time. Section headings and sub-headings are directly readable, whereas each letter of the content paragraphs is replaced by an asterisk. Clicking on a paragraph with the computer mouse causes the whole paragraph to appear in a readable form (Figure 1). The scroll bar at the right-hand side of the screen can be used to

move from page to page. The Select-the-Text software allows precise recording of reading and search patterns. Reports from previous experiments (e.g., Goldman, Saul & Coté, 1995) as well as our own pilot trials suggested that this presentation mode is easily mastered by students.

Insert Figure 1 about here

Two sets of search questions were prepared. All the questions could be answered based on textual information. The questions differed in the amount of textual information needed to provide a satisfactory answer. Low level questions (LLQ) could be answered using information found in one or two paragraphs of the text (e.g. "who measured the mass of electrons for the first time?"). High level questions (HLQ) required the students to review one or two sections of the text (e.g., what is the main difference between Rutherford and Thomson's models?). The set of questions is presented in Table 1.

Pilot study. A pilot study was conducted in order to obtain an independent estimate of the subjective importance of text elements. We asked a group of 8 psychology graduate students with some experience in text analysis to read the text and underline passages whose content "would be important to reflect in a summary". The ratings were used to compute the level of importance of the information directly targeted by the high level and low level questions, and to identify the five most important ideas which were not directly related to the search questions. The number of subjects who underlined a sentence (or part of the sentence) was considered an estimate of the sentence's level of importance. The main categories of information (gist ideas, high level target information, low level target information) along with the corresponding importance score are presented in Table 1. When one item corresponded to several sentences

from the text the average importance score of the relevant sentences was computed. A one-way ANOVA with items as a random factor, idea category as a between-item factor and the importance score as a dependent measure showed that there was a significant difference in importance across categories ($F(2,12)=9.54, p<.01$). While the gist units and high level targets did not differ significantly from each other ($F(1,8)=2.20, NS$), they obtained higher ratings than the low level targets ($F(1,8)=13.33, p<.01$ and $F(1,8)=7.94, p<.05$, respectively). Thus, gist and high level target units were both subjectively more important, or more likely to appear in a summary than the low level target units.

Insert Table 1 about here

Procedure. The main experiment was run individually. The participants were randomly assigned to the LL or HL question conditions. They were first asked to fill in a form asking for their academic background, experience with computers and other personal data. After a practice task on the presentation software, the students were asked to read the text so as to be able to use it later in order to locate the answer to a series of questions. The maximum study time was loosely set to 15-20 minutes in order to take into account differences in reading rate and study strategies. Next, the students were given the five search questions one at a time. For each question, the students were asked to search the text for relevant information in order to write a precise answer to that question. They were told to try to locate relevant information as fast as possible, while making sure that they would not miss anything important. When they felt they had finished searching, they had to click on a "pause" button and to write down their answer. The order of presentation of the questions was counterbalanced. After the series of five questions the students

completed a short filler task. Finally the students were asked to write a one-page summary of the text, as if they were to explain the text's main ideas to another student.

The main predictions were that (a) high level and low level questions would yield different search patterns. Target information for HLQs was relatively easy to locate, but quite difficult to integrate. Therefore high level questions might result in longer search episodes with longer exposure time per paragraph. Conversely, low level target information might be harder to locate, but easier to integrate. Thus, subjects might inspect more paragraphs, with a shorter exposure time per paragraph (except, maybe, for that one paragraph where the information was located). In addition we expected that reviewing the text in the HLQ condition would promote the inclusion of high level target information as well as gist information in the summary.

Results

Search strategies. Students managed to provide correct answers to the search questions in over 90% of the cases. Thus both HLQ and LLQ could be considered relatively easy. However the actual search behavior varied a lot across students and across questions. For some questions students located the target passages in a straightforward manner, whereas for other questions they browsed extensively through the text. Table 2 summarizes students' search behavior in the HLQ and LLQ groups. The total search time varied from a few seconds to several minutes, with an average of roughly 1.5 minutes per question, and did not differ significantly across groups. However, within each trial, the search time was unevenly distributed across the paragraphs that the subject decided to inspect. Subjects in the LLQ group tended to browse several paragraphs rather quickly, and to pause for a much longer time on their last paragraph opened (which, in over 80% of cases, was the target paragraph). Consequently, the review time on last selection was significantly higher in the LLQ than in the HLQ group ($t(8)=3.78, p<.01$).

Insert Table 2 here

The number of paragraphs students opened during a search trial was also highly variable (between 1 and 25 paragraphs). There was no difference in the number of different paragraphs subjects opened during one trial. However, subjects in the HLQ group tended to pause on more paragraphs during the search. Using a 4-second threshold to distinguish the paragraphs "studied" from those that were simply "browsed", we found that on 84% of the cases students in the HLQ group studied between 3 and 10 paragraphs (28% in the LLQ group). Students in the LLQ group studied 1 or 2 paragraphs in most cases (56% vs. 16% in the HLQ group). However on a few occasions (16%) they studied more than 10 paragraphs before answering the question. This was never the case in the HLQ group. These infrequent but few very high figures in the distribution of the LLQ group explained in part the non-significant result.

We counted the number of questions for which a student's search pattern was "integrative", that is, for which the student opened 3 or more paragraphs for 4 seconds or more each. The frequency of questions with an integrative search pattern was almost double in the HLQ group than in the LLQ group (4.20 vs. 2.20, $t(8)=3.24$, $p. <.02$). In other words, HLQ students engaged in an integrative kind of search almost systematically, while students in the LLQ group did so for less than half of the questions. Their search patterns on these occasions suggest that they got "lost in the text", and had to open a large number of paragraphs in order to locate the specific piece of information requested by the question. In summary, the data suggest that despite the high variability of search behavior, the HL questions promoted a deeper "review-and-integrate" strategy, whereas the LL questions triggered a "locate-and-memorize" strategy.

Summaries. Students' summaries were analyzed for length and inclusion of important ideas. The number of words per summary ranged from 156 to 232 and did not differ significantly across groups (183.6 vs. 192.4, on average in the LQ and HQ groups, respectively, $F < 1$).

Each summary was analyzed for the inclusion of key ideas (as presented in Table 1) by three scorers. Because of the high level of compression of the information in the summaries, the task of identifying key ideas in the summaries was not trivial. A scoring template with scoring criteria for each key idea was defined by one scorer and then progressively refined through trials and discussion². Then each scorer rated the 10 summaries. Inter-scorer agreement ranged from 50 to 70%, depending on the pair of scorer considered. The discrepancies were distributed across protocols and across scoring units in an unsystematic fashion. We decided to consider only those key units that had been identified by at least two of the three scorers. The scores were then converted into percentages within each category (gist idea, HL target, LL target), with 0% meaning that no unit of the category was present in the summary, and 100% meaning that the five units of the category were present.

Insert Figure 2 about here

Figure 2 shows the average percentage of units present in the protocol. A two-way ANOVA with task as a between-subject factor and type of unit as a within-subject factor showed a main effect of unit type ($F(2,16)=5.69$, $p < .05$). Subjects included more gist units than HL targets and LL targets. There was also a significant task x unit interaction ($F(2,16)=6.94$, $p < .01$). The LLQ subjects tended to include more G-units and LL-target units, whereas the HLQ subjects tended to include more HL-target units. Thus the search task did influence the type of ideas that

students included in their summaries. However, contrary to our expectation, the HL questions did not increase the inclusion of gist ideas in the summaries.

We also examined the possible influence of academic background on students' summaries. The three non-science students (1 in the LLQ group and 2 in the HLQ group) included on average less gist ideas than the seven science students (33% vs. 60%, respectively). The two non science students in the HLQ group only included one gist unit out of five in their summary (vs. 2.6 gist units in the science HLQ students). Therefore, the apparent decrease in gist unit production in the HLQ group might be due to a sampling effect.

Discussion

The primary purpose of the experiment was to demonstrate an effect of search tasks on students' comprehension of a lengthy instructional text. Students' search patterns varied as a function of question type. Students in the HLQ group tended to study more paragraphs per question (higher integration score), while students in the LLQ group open most often 1 or 2 paragraphs and concentrated on the last paragraph searched. High level questions thus promoted a "review-and-integrate" strategy, whereas the low level questions promoted a "locate-and-memorize" strategy. The "review-and-integrate" strategy consists in reviewing several paragraphs that contribute to a core topic in the text. In contrast, the "locate-and-memorize" strategy consists in browsing the text in order to locate a specific fact. Experiment 1 corroborates earlier evidence that the way students search a text depends on the type of question that they are being asked (Goldman & Durán, 1988; Vidal-Abarca et al., 1998). In addition, the data suggest that higher-level questions promote a deeper reprocessing of the text.

The type of search question also had an effect on the content of students' summaries. Students tended to include more information relevant to their search tasks in their summary of the

text. However the difference between the HLQ and the LLQ groups did not exactly follow the predicted pattern: In addition to low level targets, the LLQ students also tended to include more gist units than the HLQ students, whereas the opposite was expected. The reason for this pattern of results is not clear. Because of the small number of data points per cell and the influence of academic background, the possibility of a sampling effect may not be discarded.

Overall the results suggest that when studying a complex instructional text, the mental representation constructed during initial reading can be qualitatively modified by subsequent reprocessing of the text. In other words, the text provides a learning resource which may be utilized in various ways depending on the task demands. Search tasks that request the review and integration of key sections of the text can promote the identification of important ideas. However, the effectiveness of an appropriate task is conditional upon the success of the search episode, that is, students' ability to actually review and integrate relevant information. As a matter of fact, students' search behavior was highly heterogeneous, as evidenced by the search time and number of paragraphs opened (Table 2). On a few occasions students "got lost" in the text, opening a disproportionate number of paragraphs before they reached the answer. Thus, some students in both groups may have wasted time reviewing irrelevant sections or skimming the text for target information. A larger contrast between high- and low level search tasks may be observed if text presentation were better supportive of students' strategies.

Experiment 2

Rationale

Experiment 1 suggested that searching the text for higher-level information may have a positive influence on main idea comprehension. However, students' search behavior was highly variable, suggesting that some students had trouble locating relevant portions of the text. The

purpose of experiment 2 was to find out if making the top-level structure of the text more explicit would facilitate the search process. We hypothesized that reducing the burden of locating relevant sections of the text may provide students more opportunities to elaborate on the search questions.

Several studies have found that textual organizers, e.g., headings, paragraphing, typography and connectives facilitate main idea comprehension (Goldman, Saul & Côté, 1995; Mayer, 1984, Spyridakis & Standal, 1987). A very effective technique consists in providing the reader with some advance information concerning the contents of the text. Murray and McGlone (1997) found that the inclusion of a topic list decreased comprehension time. Lorch, Lorch and Inman (1993) presented evidence that combination of spacing, headings and overviews improved students' memory for text structure. However the effects of top-level content representations varies as a function of task setting and the readers' level of prior knowledge. Dee Lucas and Larkin (1995) asked undergraduate students to study an introductory physics text presented on a computer screen either in a traditional format (i.e., with headings and subheadings embedded in the text), in a hypertext format with an alphabetical index, or in a hypertext format with a hierarchical overview (i.e., a tree diagram representing the main concepts to be learned). When students were assigned a general comprehension task, the hierarchical overview increased students' reviewing of the text and improved their memory for main ideas. However when students were asked to read with the purpose of summarizing the text, no advantage was found for the hierarchical overview. The summary task may have focused students' attention on the global organization of ideas in the text, overshadowing the effect of the hierarchical overview. Hofman and van Oostendorp (1999) observed that a graphical overview was detrimental to low prior knowledge students' deep comprehension of a scientific text, compared to a topic list. They

suggested that the overview hindered low knowledge students' comprehension because it drew their attention to the textual macrostructure at the expense of the microstructure. It is worth noticing that contrary to the simple hierarchical diagram used by Dee Lucas and Larkin, Hofman and van Oostendorp's graphical organizer was a rather complex network diagram, with no clear orientation and several types of links whose meaning was only suggested. Its effectiveness may have also been challenged by its intrinsic complexity.

Thus, even though the majority of studies have found positive effects of structural overviews and other textual organizers, there seem to be some conditions to their effectiveness. One of the conditions is that the content representation must be clear and non ambiguous. Another condition is that study directions do not explicitly focus students' attention on the organization of the text.

We hypothesized that in a simple structural overview (e.g., a hierarchical table of contents) would increase students' reviewing of the text (Dee Lucas & Larkin, 1995). Furthermore, the overview would facilitate the selection of a relevant text passage (Mosenthal, 1990). The overview may provide cues useful both for the high level and low level questions. Consequently, students would spend less time matching the question focus (Graesser & Franklin, 1990) with text contents, regardless of question type.

If students spend less time accessing relevant information, then they will have more time to actually process it. In the case of high level questions, students would be able to devote more time to the review and integration of relevant main ideas. Thus, in the high level question condition, the presence of a structured overview might increase the inclusion of gist information in students' summaries. The result would be an interaction between question type and overview, with the best summaries being written in the high level question/structured overview condition.

Method

Subjects. 28 University of Poitiers students (13 females, 15 males) between the ages of 20 and 25 participated as paid volunteers. All the participants were native French speakers. In order to control the effects of academic background, only science students were recruited. 15 of the participants (9 females, 6 males) were physics or chemistry undergraduates, the others were enrolled in various scientific or engineering programs.

Materials. The contents of the text were identical to those used in Experiment 1. A new version of the text was created in which page 1 presented the title of the text plus an overview (explicit overview condition). In the control condition, page 1 only presented the title (implicit overview condition). We used a set of questions drawn from experiment 1, with some changes in phrasing in order to make the location of target information less obvious for some items.

Design and procedure. The factors "type of question" (high level or HL vs. low level or LL) and "overview" (explicit or implicit) were combined so as to obtain four between-subject experimental conditions: HL/explicit, HL/implicit, LL/explicit and LL/implicit. The procedure was similar to experiment 1: The participants were randomly assigned to one of the four conditions. They were trained with a practice text and two practice questions. They were asked to study the critical text for 15 to 20 minutes. Then they completed five search tasks, a filler task, and finally they wrote a one-page summary of the text.

In addition, a surprise memory task was added after the summary task, in order to evaluate students' memory for the text structure. The students were given 12 sentences extracted from the text and presented in a random order. There was roughly one sentence per subsection of the text. The sentences' level of importance, as estimated in the pilot study, ranged between 0 and 5 (i.e., none of the sentences corresponded to a gist idea). Three of the sentences were part of the target

passages for high level questions; 2 were targets for low level questions, 1 was a target for both types of questions and 6 were not part of a target passage. The students were provided with a copy of the overview and asked to recall in what section of the text the information was originally presented. The purpose of this task was to assess memory for the text structure independent from students' writing skills.

The main hypotheses were as follow:

- Study and search strategies: High level and low level questions will promote review-and-integrate and locate-and-memorize search patterns, respectively; the overview will speed up the initial study of the text and facilitate the search process, especially for high level questions.
- Text comprehension and memory for main ideas: Students will tend to include in their summary the type of information targeted by the search questions; due to the review-and-integrate strategy, students in the HLQ groups will include more gist ideas in their summaries and recall the location of target sentences more accurately.

Results

Initial study phase. The total time spent studying the text and the number of paragraphs reviewed during the initial study phase are shown in Table 3. The time spent on the cover page (i.e., title or the title plus overview, depending on the condition) was excluded from the analysis so that the data were strictly comparable across conditions.

Insert Table 3 about here

The explicit overview increased the average study time by about 20%. However, due to the large within-group variances the difference was marginally significant ($F(1,24)=3.80, p<.10$).

The increase in study time was partly due to a larger proportion of paragraphs read more than once. The percentage of paragraphs re-read was 55% with an explicit overview versus 15% without ($F(1,24)=7.23$, $p<.05$). Thus, the explicit overview acted as an advance organizer, pushing students to review greater portions of the text. In fact, the distribution of re-reading was fairly bimodal: 71% of the students reviewed either less than 10% or more than 90% of the paragraphs. Seven out of the eight students in the latter category were in the explicit overview condition.

Search phase. Search strategies were analyzed based on the same four parameters as in experiment 1 (search time per question, review time of last paragraph searched, number of paragraphs opened and number of integrative search patterns). The results are presented in Table 4.

Insert Table 4 about here

The results were analyzed using two-way ANOVAs with type of question and overview as between-subject factors. There was a nonsignificant 14% decrease in search time in the explicit overview condition ($F<1$). The review time of the last paragraph selected was longer in the LLQ condition than in the HLQ condition ($F(1,24)=6.25$, $p<.05$), and was unaffected by the overview. The number of paragraphs opened was smaller in the condition with an explicit overview ($F(1,24)=9.94$, $p<.05$), and remained unaffected by question type ($F<1$). Finally, students engaged in an integrative search more often in the HLQ than in the LLQ condition ($F(1,24)=17.47$, $p<.01$). The implicit overview also significantly increased the frequency of

integrative searches ($F(1,24)=5.85, p<.05$). There was no interaction between the two factors ($F<1$).

Overall, the results were very similar to those of experiment 1. We replicated the effects of question type on the search patterns (duration of last review episode, and frequency of integrative search patterns). In addition, we found an effect of the overview on students' search behavior. The explicit overview decreased the number of paragraphs studied, as well as the frequency of integrative search episodes. These results support the existence of review-and-integrate vs. locate-and-memorize search patterns, and suggest that text search is facilitated by an explicit overview.

Content analysis of the summaries. Students' summaries were analyzed using the same procedure as in experiment 1 (Figure 3).

Insert Figure 3 about here

A three-way ANOVA with task and overview as between-subject factors and type of unit as within-subject factor was performed. Again, the summaries included more gist units than HL and LL targets ($F(2,48)=29.67, p<.01$). There was a significant interaction between the type of question and the type of units included in the summaries ($F(2, 48)=5.61, p<.01$). Partial comparisons showed that the HLQ group included a larger proportion of HL targets ($F(1, 24)=5.42, p<.05$), while the LLQ group included more LL targets ($F(1, 24)=5.24, p<.05$). There was no significant difference in the proportion of gist units included in the two groups ($F(1,24)=1.63, NS$). No other effect or interaction was significant.

The results basically replicated those of experiment 1. The students tended to include question-relevant information in their summary. However the number of gist units included was similar in the two groups. The explicit overview had no effect on the summaries.

Memory for text structure. The number of sentence locations correctly recalled was computed and subjected to a two-way ANOVA with task and overview as between-subject factors (Figure 4). The HLQ group recalled on average a larger number of sentence locations than the LLQ group ($F(1, 24) = 5.61, p < .05$). There was no effect of the type of overview and no interaction between the two factors.

Insert Figure 4 about here

Thus, the task performed after reading the text did have an influence on the subjects' memory for the text's top-level structure. Students in the HLQ condition had a more accurate memory for the location of content information. More surprisingly, the presence of an explicit overview did not have any influence on students' performance.

General discussion

The two experiments showed an effect of search tasks on both the processes and the products of text comprehension. High level questions resulted in a review-and-integrate type of search, as evidenced by more search episodes with both several paragraphs opened and a longer study time per paragraph. In contrast, low level questions triggered search episodes with very few paragraphs opened, and a long pause on the last paragraph opened. On a few occasions students in the LLQ group had to browse through many paragraphs before they located the target information. This suggests a "locate and memorize" strategy in which the students try to extract a

specific piece of information and then to memorize literally before they answer. These search patterns are similar to those obtained by Vidal-Abarca et al. (1998).

The type of search task also influenced students' summaries: Even though the summarization task was distinct from the search task and presented after a filler task, students in each group tended to include information relevant to their search task in the summary.

Experiment 2 showed that the high level search task improved students' memory for the text structure. In addition, experiment 2 showed that an explicit overview encouraged students to review the text during the initial study phase. The explicit overview also tended to reduce the search time and the number of paragraphs searched. Both the average search time and the number of paragraphs opened per questions were higher in experiment 2 than in experiment 1. This may be easily explained by the rephrasing of some questions which made the localization of information harder. Despite this difference the pattern of effects was very similar in the two experiments.

Interestingly, searching for high level information did not take more time than searching for details. However, the data suggest that the activities which take place during high vs. low level searches are fundamentally different. Low level questions focus on a single piece of information, e.g., a number or a fact (Graesser & Franklin, 1990). The search can be conducted on the basis of shallow semantic processing, e.g., word recognition. The task is time consuming only because the location of low level information is harder to guess or to recall. Consequently, subjects spend most of their search time eliminating irrelevant sections of the text. In the case of higher-level questions, the student must review and integrate whole sections of the text (Andre, 1979; Mosenthal, 1996). Moreover the student must synthesize information, that is, condense the raw information into main points so as to build up an answer. Online records of students' search

patterns showed that they used different search strategies when asked high vs. low level questions. The low level questions prompted a locate-and-memorize strategy, i.e. the students quickly browsed the text and studied the target segment, once located, for a longer time. The high level questions prompted a review-and-integrate strategy, where the students went forth and back between the question's target passages. We suggest that the review-and-integrate strategy supported the construction of a structured representation of the main features of the atomic models as well as their similarities and differences, i.e., a situation model of the text (van Dijk & Kintsch, 1983).

Our expectation was that high level questions would improve students' identification and memory for main points, and hence the quality of their summaries. We found partial evidence in favor of that claim. Students in the HLQ group did include more HLQ-relevant information in their summaries, that is, more important information besides gist ideas. The ideas included by subjects in the LLQ group were structurally less important (Table 1). The high recall rate of gist ideas (about 70% in study 2) suggests that the participants had efficient summarization strategies. This may explain why the high level search tasks only increased the inclusion of question-relevant concepts. However, because we did not include a "no-question" condition, it is difficult to evaluate the impact of search tasks in absolute terms. We can only reason based on the comparison of high level and low level search tasks.

The explicit overview had a strong impact on the participants' study behavior. They read the text for a longer period of time and reviewed more paragraphs when the text was presented with an explicit overview. These results are consistent with previous studies of reading electronic texts (Dee Lucas & Larkin, 1995). Clearly the overview acted as an advance organizer which informed the students about the length, structure and contents of the text. The explicit overview

also facilitated the search process, possibly in part because of the more careful initial reading of the text. However, the explicit overview had very little impact on the summaries or on the recall of the text headings. The latter finding adds to a growing body of mixed results concerning the effects of structured overviews on deep comprehension (Hofman & van Oostendorp, 1999; Stanton, 1994). Even though a lack of difference does not prove anything per se, some tentative post-hoc explanations may be briefly suggested: First, the lack of explicit overview may have forced students to pay more attention to the subheadings inserted in the text during both the initial reading phase and during the search phase. This would account for the longer study time observed in the implicit overview group, as well as for the similar performance on the post-tests. An alternate explanation is that gist ideas (e.g., "Thomson introduced a new atomic model") were so simple that science majors could learn them by simply reading the text and paying attention to sub headings. In other words, the explicit overview was effective as a selection tool (Mosenthal, 1996), but not as a learning device. Partial support for this explanation is provided by Vidal-Abarca, Gilabert and Rouet's (1998) findings that high level questions did not benefit students with high prior knowledge of the topic. However, further experiments are needed in order to find out under which conditions explicit overviews will have a facilitative effect on deep comprehension.

While our results are consistent with previous studies, their generalization is restricted by the particular presentation mode used in our experiments, i.e., a computerized display with a special masking procedure. It is known that reading from a computer screen is generally slower and less accurate than reading from paper (Dillon 1994). Moreover, special presentation features such as short page length or masked text may also influence students' strategies. Because we did not include a control paper condition, little can be said about the size and direction of these

effects. However, given that computers are increasingly used as study devices, we believe that our data are nevertheless of some practical interest.

Finally, our study stresses the importance of task contexts in learning from instructional texts. As Guthrie & Kirsch (1987) pointed out, our understanding of comprehension processes has been so far somewhat limited because research studies have typically focused on short, simple passages that can be memorized even in the absence of a specific task context. For such basic comprehension tasks, the hierarchy of information in students' representation can be accurately predicted on the basis of structural importance (Kintsch & van Dijk, 1978). However in naturalistic contexts people are frequently confronted with longer and more complex texts, which require a much more selective comprehension process. In such situations the task context, or what the text is being used for, becomes a strong criterion to determine what the important information is. Searching a text for particular pieces of information induces processes complementary to the comprehension processes that occur during initial, linear reading (Einstein, McDaniel, Owen & Côté, 1990). The extra-processing brought by the search task makes target information more salient in the representation, and more likely to be included in a summary.

We suggest that locating information and reading comprehension are two aspects of a more general category which may be called "information tasks", or tasks which require individuals to use an external information source in order to answer a question or to solve a problem. Information tasks are distributed along a "goal specificity" continuum, with fact localization and text comprehension at the extremes, and most learning tasks involving instructional texts and documents somewhere in the middle. Knowing when and how to use a text in order to answer a question is a skill that can be trained through direct instruction (Raphael, 1984). Assuming that the learner is aware of task demands, some tasks (e.g., the high level

search tasks in the present experiments) promote reviewing and integration processes that facilitate deep comprehension. When using texts for instructional purposes, such tasks may be used to scaffold students' comprehension strategies.

Authors' notes

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Table 1: Gist ideas and target information (abridged) for high level and low level search questions (experiment 1) along with the importance level of respective passages.

1. Gist ideas (G)		importance level (max=8)
G1	Atomic models were created to explain natural phenomena.	7.00
G2	The first model was proposed by Dalton.	7.00
G3	Some experiments contradicted Dalton's model.	6.25
G4	Thomson introduced a new atomic model.	7.00
G5	Rutherford's atom was made of a small kernel with electrons gravitating around it.	8.00
2. Target ideas for high level search questions (HLQ)		importance score
HLQ1 (What are the characteristics of Dalton's atom?)		
	Dalton's atom was like a compact sphere (...).	7.50
HLQ2 (What is an atom made of according to Thomson?)		
	Thomson's atom involved a soft sphere with electrons incrustated in it.	5.80
HLQ3 (What was an essential difference between Dalton's and Thomson's models of the atom?)		
	Thomson's atom was divisible while Dalton's atom was not.	6.29
HLQ4 (In the experiment by Rutherford what were the trajectories of the particles?)		
	In Rutherford's experiment particles followed several trajectories (...).	6.00

HLQ5. (What is the main difference between Thomson's and Rutherford's models?)

Rutherford's atom was basically empty whereas Thomson's was not.

6.63

(Table 1 continues next page)

Table 1 (continued)

3. Target ideas for low level search questions (LLQ)	importance score
LLQ1 (What is stated in the Law of Constant Proportions)	
Composition of a compound can be expressed as a ratio of integers.	3.50
LLQ2. (Who measured the mass of electrons for the first time?)	
Thomson was the first scientist to measure the mass of electrons.	4.00
LLQ3. (What is the phenomenon now know as radioactivity?)	
Radioactivity is when a substance emits particles.	4.00
LLQ4.(What particles were projected on the gold leaf in Rutherford's experiment?)	
Alpha particles were projected on a gold leaf.	7.00
LLQ5. (According to Rutherford what is the ratio between the diameter of an atom and the diameter of its kernel?)	
An atom's kernel is 10000 to 100000 times smaller than the atom.	3.00

Table 2: Parameters of students' search behavior as a function of question type (experiment 1).

	Type of questions		
Measures	<u>High level</u>	<u>Low level</u>	<u>p.</u>
Search time per question (seconds)			
<u>M</u>	80	91	NS
<u>SD</u>	38	8	
Review time on last paragraph selected (seconds)			
<u>M</u>	19	33	.01
<u>SD</u>	5	8	
Paragraphs opened per question			
<u>M</u>	5.3	5.0	NS
<u>SD</u>	1.8	0.7	
Integrative search patterns (max=5)			
<u>M</u>	4.2	2.2	
<u>SD</u>	1.1	0.8	.02

Note. Integrative search patterns were defined as questions for which subjects paused on 3 paragraphs or more.

Table 3: Initial study time and percent paragraphs reviewed as a function of question type and overview (experiment 2).

	Questions			
	High level		Low level	
	<u>M</u>	<u>SD</u>	<u>M</u>	<u>SD</u>
Initial study time (seconds)				
Explicit overview	1118	268	1177	423
Implicit overview	933	224	914	262
Percent paragraphs reviewed				
Explicit overview	51.3	47.1	59.5	49.0
Implicit overview	11.2	17.1	19.3	36.2

Note. Data are presented separately for high level and low level question conditions only for the sake of clarity. These conditions were ineffective during the initial reading of the text.

Table 4: Parameters of search behavior as a function of question type and overview (experiment 2)

	Questions			
	High level		Low level	
	<u>M</u>	<u>SD</u>	<u>M</u>	<u>SD</u>
Search time per question (seconds)				
Explicit overview	112	66	109	68
Implicit overview	127	40	125	54
Review time on last selection				
Explicit overview	21.8	8.8	39.6	34.0
Implicit overview	17.2	7.2	33.7	5.5
Paragraphs opened per question				
Explicit overview	5.60	1.45	4.40	2.18
Implicit overview	7.26	1.21	7.86	3.18
Integrative search patterns (max. 5)				
Explicit overview	4.29	0.95	2.86	0.69
Implicit overview	5.00	0.00	3.71	1.25

Figure captions

Figure 1: Screenshot of the Select-the-Text software showing a passage of the atoms text. The paragraph just below the heading "Le modèle de Dalton" (Dalton's model) has been selected using the mouse and appears in a readable form.

Figure 2: Types of units included in the summaries as a function of the type of search task (experiment 1).

Figure 3: Types of units included in the summaries as a function of the type of search task (experiment 2).

Figure 4: Memory for text structure as a function of question type and overview.

¹ We thank Béatrice Cossais for her assistance in translating the text from Spanish.

² The scoring criteria (in French) are available upon request to the first author.